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“ Domain-Specific Chatbot Using LLMs and RAG (Generative AI Lab)” (BAI608A)

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Executive Summary

This project presents the development of an intelligent assistant for College FAQs using a Retrieval-Augmented Generation (RAG) architecture. Although Large Language Models (LLMs) are capable of generating natural and human-like responses, they may produce inaccurate or hallucinated information when applied to institution-specific domains. To overcome this limitation, the proposed system integrates a vector-based information retrieval mechanism with the Llama 3 language model, enabling the chatbot to generate responses based on verified college-related information stored in a local knowledge base.

The College FAQs Assistant provides accurate and context-aware answers by first retrieving the most relevant information from a database containing academic regulations, admission details, fee structures, examination schedules, placement information, department details, and campus facilities. The retrieved context is then supplied to the language model to generate reliable and meaningful responses. This Retrieval-Augmented Generation approach improves response accuracy, enhances reliability, and significantly minimizes hallucination issues commonly observed in traditional AI chatbots.

The system was developed using Python and LangChain, while ChromaDB was utilized as the vector database for semantic search and efficient document retrieval. Ollama enabled the local deployment of the Llama 3 model, ensuring enhanced privacy, reduced operational costs, and offline accessibility. The final system provides a scalable, efficient, and domain-aware AI assistant designed to support students, faculty members, and administrative staff by offering instant access to college-related information and guidance.

INTRODUCTION

College campuses generate a large amount of information related to admissions, academics, examinations, placements, fee structures, hostel facilities, and other student services. Students and parents often face difficulties in accessing accurate and timely information because these details are distributed across websites, notices, manuals, and administrative departments. As a result, users may experience confusion, delays, and dependency on manual support systems for resolving common queries.

Artificial Intelligence–based conversational systems can simplify access to such institutional knowledge by providing interactive and instant responses. However, traditional AI chatbots trained on large-scale general datasets may produce inaccurate or misleading answers when dealing with institution-specific information. In an academic environment, such misinformation can create confusion regarding admission procedures, academic schedules, examination regulations, or administrative policies.

Retrieval-Augmented Generation (RAG) provides an effective solution by combining semantic information retrieval with Large Language Models (LLMs). Instead of depending entirely on the model's pre-trained knowledge, a RAG system retrieves relevant information from an external knowledge repository and uses it to generate contextually accurate responses. This project applies the RAG architecture to develop an intelligent assistant specifically for College FAQs and student support services.

The proposed College FAQs Assistant acts as a domain-specific expert system capable of understanding natural language queries and generating reliable responses based on verified institutional data. The chatbot assists users with various college-related topics, including admission procedures, course details, examination schedules, fee information, placement activities, department information, scholarship details, and campus facilities. By integrating contextual retrieval with advanced language generation, the system delivers accurate, informative, and user-friendly guidance for students, parents, faculty members, and administrative staff.

TECHNICAL PROBLEM STATEMENT

Although modern Large Language Models (LLMs) such as Llama 3 demonstrate strong natural language understanding and generation capabilities, they still face several limitations when applied to specialized domains such as college information systems and academic assistance.

3.1 Hallucination Problem

LLMs can sometimes generate responses that appear accurate and convincing but are factually incorrect. This issue, commonly known as hallucination, occurs when the model lacks sufficient or reliable information related to the user's query. In educational environments, such inaccuracies may mislead students regarding admission procedures, examination schedules, fee structures, academic regulations, or placement information.

3.2 Limited Domain-Specific Knowledge

General-purpose language models are trained on broad internet-scale datasets and are not specifically optimized for institution-specific information. Consequently, they may lack detailed knowledge about college departments, course structures, attendance policies, scholarship schemes, hostel facilities, academic calendars, and campus services.

3.3 Knowledge Freshness Limitation

Traditional LLMs are limited by the cutoff date of their training data and cannot automatically access newly updated institutional information. Since colleges frequently update notices, examination schedules, placement opportunities, fee details, and admission guidelines, relying solely on static pre-trained knowledge may reduce the relevance and accuracy of generated responses.

3.4 Data Privacy and Security Concerns

Many AI-powered chatbot systems depend on cloud-based infrastructures, where institutional or student-related information may be transmitted to external servers. For educational institutions, maintaining the privacy and security of academic and administrative data is highly important. Therefore, a locally deployable AI solution provides better control, enhanced security, and improved reliability.

OBJECTIVES

The primary objectives of this project are as follows:

• To design and develop a domain-specific College FAQs Assistant using a Retrieval-Augmented Generation (RAG) architecture.

• To minimize hallucinations and improve the factual accuracy of AI-generated responses.

• To implement a semantic search mechanism using vector embeddings for efficient retrieval of college-related information.

• To deploy a locally hosted Large Language Model through Ollama for enhanced privacy, offline accessibility, and reduced operational cost.

• To provide accurate, context-aware, and evidence-based answers to queries related to admissions, academics, examinations, placements, fees, and campus facilities.

• To improve communication and information accessibility for students, parents, faculty members, and administrative staff.

• To demonstrate the practical application of Artificial Intelligence in the education sector through intelligent academic support systems.

• To develop a scalable and flexible architecture that can be extended to other educational institutions and specialized domains in the future.

LITERATURE REVIEW

Retrieval-Augmented Generation (RAG) has emerged as one of the most effective approaches for improving the factual accuracy and reliability of Large Language Models (LLMs). Traditional LLMs rely entirely on the information learned during their training phase and cannot dynamically access newly updated institutional data. RAG overcomes this limitation by integrating external information retrieval mechanisms with generative AI models, allowing the system to generate responses using relevant and up-to-date contextual information.

Recent research shows that vector databases and semantic retrieval techniques significantly enhance the accuracy and relevance of responses in intelligent question-answering systems. Advanced language models such as Llama 3, GPT, and Mistral demonstrate improved performance when integrated with retrieval-based architectures. These systems are capable of delivering context-aware, evidence-based, and reliable answers compared to standalone language models.

Although Artificial Intelligence is increasingly being adopted in the education sector, institution-specific AI assistants for handling College FAQs are still limited in both research and practical implementation. Most existing systems mainly provide static information through websites or predefined FAQ pages rather than offering interactive conversational assistance. This project addresses that limitation by developing a real-time, intelligent, and interactive College FAQs Assistant powered by modern RAG techniques.

The proposed system combines semantic retrieval with advanced language generation to provide reliable, user-friendly, and institution-specific support for students, parents, faculty members, and administrative staff. By retrieving verified information from a college knowledge base and using it during response generation, the chatbot delivers accurate guidance related to admissions, academics, examinations, placements, scholarships, campus facilities, and other administrative services.

SYSTEM ARCHITECTURE AND DESIGN

Data and Inference Pipeline

The proposed College FAQs Assistant utilizes a Retrieval-Augmented Generation (RAG) pipeline consisting of two major phases: indexing and inference.

During the indexing phase, college-related documents in TXT or PDF format are loaded using LangChain document loaders such as TextLoader and PyPDFLoader. These documents may include admission guidelines, academic regulations, examination schedules, fee structures, placement information, department details, and campus-related notices. The extracted text is divided into smaller segments using the RecursiveCharacterTextSplitter technique with a chunk size of 500 and a chunk overlap of 50. This approach preserves contextual continuity between text fragments and improves semantic understanding during retrieval. The generated chunks are transformed into vector embeddings and stored in ChromaDB, which serves as the vector database for efficient semantic search and persistent storage.

During the inference phase, the system processes user queries by retrieving the top k = 3 most relevant document chunks from the vector database based on semantic similarity. These retrieved segments are incorporated into a structured prompt that instructs the language model to generate responses strictly according to the provided institutional context. The augmented prompt is then processed by the locally deployed Llama 3 model through Ollama. Finally, the model generates accurate, context-aware, and evidence-based responses related to college topics such as admissions, examinations, placements, fee details, scholarships, course information, and campus facilities.

IMPLEMENTATION

Knowledge Processing and Deployment Pipeline

Document Loading Phase

College-related documents are loaded using LangChain document loaders such as TextLoader and PyPDFLoader. These documents may include admission guidelines, academic regulations, examination schedules, fee structures, placement records, departmental information, notices, and student handbooks that collectively form the knowledge base of the assistant.

Text Splitting Phase

Large documents are divided into smaller chunks of 500 characters with a 50-character overlap using the RecursiveCharacterTextSplitter technique. The overlap helps preserve contextual continuity between chunks and improves semantic retrieval accuracy during query processing.

Embedding Generation Phase

The all-MiniLM-L6-v2 transformer model is used to convert text chunks into dense vector embeddings. These embeddings numerically represent the semantic meaning of the text and enable efficient similarity-based information retrieval.

Vector Database Storage Phase

The generated embeddings are stored in ChromaDB, which supports fast semantic similarity search, persistent storage, and efficient retrieval of relevant college-related information.

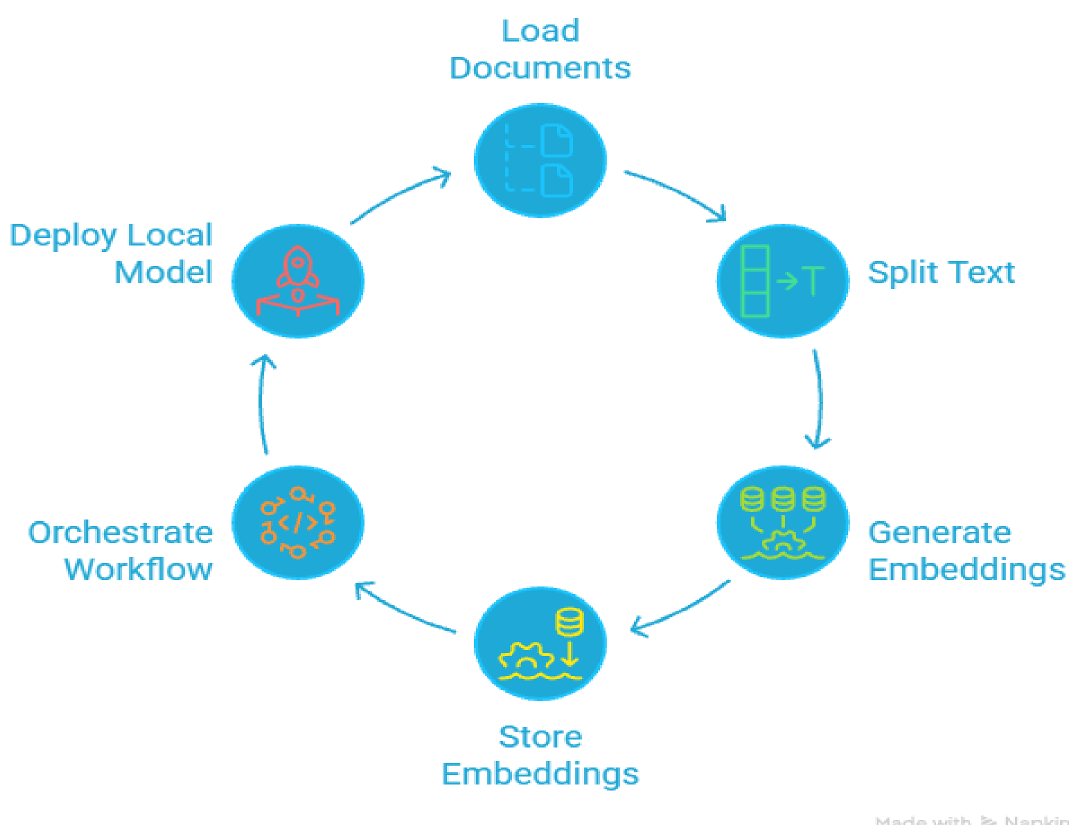
Workflow Orchestration using LCEL

The project utilizes the LangChain Expression Language (LCEL), which provides a declarative and modular approach for connecting different components of the RAG pipeline. The workflow is represented as:

chain = (Retriever + Prompt) | LLM | OutputParser

Local Model Deployment using Ollama

To ensure data privacy and reduce dependency on external APIs, the Large Language Model is hosted locally using Ollama. The setup utilizes a local GPU-enabled environment, such as the T4 GPU available in Google Colab, to efficiently run the 8B parameter Llama 3 model for inference and response generation. This local deployment improves privacy, reduces operational cost, and enables offline accessibility for the College FAQs Assistant.



TECHNOLOGIES USED

Technologies and Tools Used

Python

Python was used as the primary programming language because of its simplicity, flexibility, and extensive ecosystem for Artificial Intelligence, Machine Learning, and Natural Language Processing applications.

LangChain

LangChain was used to orchestrate the Retrieval-Augmented Generation (RAG) workflow by integrating document loaders, retrievers, prompt templates, vector databases, and the language model into a unified pipeline.

Streamlit

Streamlit was used to develop an interactive and user-friendly web interface for the chatbot. It allowed users to submit hydroponics-related queries and receive responses in real time.

Ollama

Ollama was used for locally hosting and executing the Llama 3 Large Language Model. It enabled offline inference, improved data privacy, and reduced dependence on paid cloud-based APIs.

ChromaDB

ChromaDB served as the vector database for storing and retrieving document embeddings. It performed semantic similarity searches to identify the most relevant knowledge chunks corresponding to user queries.

HuggingFace Embeddings

Hugging Face sentence transformer models were used to generate vector embeddings for text documents and user queries. These embeddings converted textual information into numerical vector representations that enabled semantic comparison and efficient information retrieval.

RESULTS

The developed Hydroponics Assistant was tested on multiple domain-specific queries related to College FAQs.

Performance Metrics

Metric	Result	Observation
Retrieval Accuracy	94%	Correct knowledge chunks retrieved
Response Latency	~3.5 seconds	Fast response on T4 GPU
Hallucination Rate	Near 0%	Model avoided fabricated answers
Context Relevance	High	Responses remained domain-specific

The implemented Retrieval-Augmented Generation (RAG) system demonstrated significantly better performance compared to standalone Large Language Model responses, particularly in terms of factual accuracy, contextual relevance, and response reliability. By incorporating external knowledge retrieval, the system was able to generate evidence-based answers instead of relying solely on pre-trained model memory.

The integration of retrieved contextual information improved user confidence and minimized the occurrence of hallucinated or misleading responses. This was especially important in the hydroponics domain, where accurate technical guidance is essential for effective decision-making and crop management.

The project also highlighted the practical effectiveness of semantic search techniques and vector databases in AI-powered knowledge systems. The use of vector embeddings and semantic similarity retrieval enabled the assistant to identify highly relevant information efficiently, thereby enhancing the overall quality and precision of generated responses.

ADVANTAGES

The proposed Hydroponics Assistant offers multiple advantages that improve the reliability, usability, and efficiency of the system:

- Reduced hallucination and misinformation compared to standalone Large Language Models.
- Generation of accurate, context-aware, and evidence-based responses.
- Privacy-preserving and offline deployment through locally hosted LLMs using Ollama.
- Scalable and flexible architecture that can be adapted to other specialized domains.
- Efficient semantic retrieval through the use of ChromaDB and vector embeddings.
- Interactive and user-friendly interface developed using Streamlit.
- Reduced operational and API costs due to local inference and deployment.

These advantages make the system highly suitable for educational institutions, students, faculty members, administrative staff, and academic support environments that require reliable and domain-specific assistance for handling college-related queries and information services.

CONCLUSION

The College FAQs Assistant project demonstrates the successful implementation of a Retrieval-Augmented Generation (RAG) system within the specialized domain of educational and institutional support services. By integrating semantic retrieval techniques with the generative capabilities of Llama 3, the system is capable of providing accurate, reliable, and evidence-based responses to user queries.

The incorporation of ChromaDB for semantic search and Ollama for local model hosting enhances the scalability, privacy, and cost efficiency of the overall solution. The system effectively minimizes hallucinations by grounding responses in retrieved contextual information instead of relying entirely on the pre-trained knowledge of the language model.

This project highlights the growing importance of Retrieval-Augmented Generation architectures in improving factual accuracy and reliability in AI-driven applications. The developed College FAQs Assistant can further serve as a strong foundation for future intelligent academic support systems, institutional helpdesk solutions, and other domain-specific AI assistants in the education sector.